

1. Administrative & Study Identification

Full Study Title: Phase 1, Open Label, Single Dose Study to Investigate the Effect of Renal Impairment on the Pharmacokinetics of Enpatoran in Male and Female Participants
Short Title/Acronym: Effect of Renal Impairment on Enpatoran Pharmacokinetics
Protocol Number: MS200569_0051
NCT Number: NCT06589713
Sponsor Name: Merck Healthcare KGaA, Darmstadt, Germany
Investigational Product: Enpatoran (M5049) - TLR7/8 Inhibitor
Phase of Development: Phase 1
Medical Monitor: Medical Monitor / Principal Investigator at Merck Healthcare KGaA

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the effect of renal impairment on the pharmacokinetics (PK) of enpatoran.
Secondary Objectives: To evaluate the safety and tolerability of enpatoran, including the incidence of treatment-emergent adverse events (TEAEs), and changes in vital signs, clinical laboratory parameters, and electrocardiograms (ECGs) up to Day 35.

2.2 Background & Rationale Enpatoran is an oral small-molecule TLR7/8 inhibitor under clinical development for autoimmune conditions such as systemic and cutaneous lupus erythematosus. As part of the clinical pharmacology characterization, it is vital to determine whether impaired kidney function significantly alters the drug's clearance and overall exposure. This study provides necessary data to guide dose adjustments and optimize the safety profile for patients presenting with comorbid chronic kidney disease (CKD).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional
Control Type: Active matched parallel control (Group 1: Normal Renal Function vs. Group 2: Impaired Renal Function)
Blinding: Open-label
Randomization: Non-Randomized

3.2 Planned Enrollment & Duration Target \$N\$: Not explicitly specified (standard Phase 1 small cohort assignment for men and women aged 18 to 80 years)
Number of Sites: Multicenter (e.g., CRS Clinical Research

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female, aged 18 to 80 years. 2. **For Control Group 1 (Normal Renal Function):** Medical history and physical examination without any ongoing clinically relevant findings; absolute lymphocyte count and absolute neutrophil count must be within normal limits. 3. **For Group 2 (Impaired Renal Function):** Chronic kidney disease (CKD) with stable medication for at least 1 month prior to screening; no ongoing clinically relevant findings except for the underlying disease leading to renal impairment.

4.2 Exclusion Criteria 1. History or presence of respiratory, gastrointestinal (including bariatric or other gastric surgeries), hepatic (including hepatorenal syndrome), cardiovascular (such as heart failure, arrhythmia), psychiatric, or neurological disorders (e.g., epilepsy). 2. History of drug hypersensitivity or serious allergic reactions leading to hospitalization. 3. Prior history of cholecystectomy or splenectomy, severe head trauma within 6 months, or severe depression within 12 months.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose given once. **Route of Administration:** Oral (enpatoran tablets). **Duration of Treatment:** Single-dose administration followed by safety and PK follow-up up to Day 35.

5.2 Concomitant & Prohibited Therapy Permitted: Stable medication for underlying diseases leading to renal impairment (must be stable for \$\\ge 1\$ month prior). **Prohibited:** Any therapies that may affect drug absorption or significantly interfere with the objectives of the study/PK clearance evaluations.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Days -28 to -1	Informed Consent, Medical History, Vital Signs, Labs, Renal Function Assessment
2	Day 1	Single Oral Dose of Enpatoran, Intensive PK blood sampling
3	Day 1 to Day 3	Intensive PK sampling at predose, 0.5, 1, 1.5, 2, 4, 8, 12, 24, 36, and 48 hours postdose
4	Up to Day 35	Final Safety Labs, ECG, Vital Signs, TEAE Evaluation

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Pharmacokinetic (PK) Plasma Concentrations of Enpatoran. **Measurement Method:** Blood sample analysis for PK parameters measured at predose, 0.5, 1, 1.5, 2, 4, 8, 12, 24, 36, and 48 hours postdose.

7.2 Secondary & Exploratory Endpoints * Number of Participants with Treatment-Emergent Adverse Events (TEAEs) up to Day 35. * Number of Participants with Abnormal Vital Signs, Laboratory Variables, and 12-Lead Electrocardiogram (ECG) up to Day 35.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of TEAEs via clinical site monitoring, laboratory tests, and ECG findings.

Serious Adverse Events (SAEs): Routine Phase 1 immediate reporting (typically 24 hours). **Data Monitoring Committee (DMC):** Internal safety monitoring managed by the sponsor (Merck Healthcare KGaA).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Set (subjects providing evaluable PK plasma samples); Safety Set (subjects receiving the dose of enpatoran). **Sample Size Justification:** Sized adequately for a standard Phase 1 PK trial characterization to establish the ratio of exposure parameters (AUC, Cmax) between subjects with renal impairment and normal matched controls.

Handling of Missing Data: Standard Phase 1 PK modeling rules apply (e.g., values below the limit of quantification treated as zero for certain metrics).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: On-site and remote monitoring to ensure protocol adherence, correct patient categorization by renal function, and rigorous PK sampling times. **Audit Trail:** Standard 21 CFR Part 11 compliant EDC systems, eCRF locking, and rigorous quality control for the bioanalytical laboratory assessing enpatoran concentrations.

11. Study Identifiers & Governance

Primary ID: MS200569_0051

Registry Number: NCT06589713

Sponsor/Lead Organization: Merck Healthcare KGaA

Study Title: Effect of Renal Impairment on Enpatoran
Pharmacokinetics

Official Regulatory Title: Phase 1, Open Label, Single Dose Study
to Investigate the Effect of Renal Impairment on the
Pharmacokinetics of Enpatoran in Male and Female Participants

12. Clinical Framework (Enpatoran Specific)

Primary Condition: Renal Impairment

Diagnosis Threshold: Clinically defined Chronic Kidney Disease
(for Group 2) vs. Normal Renal Function (for Group 1)

Medical Methodology: Pharmacokinetic evaluation of TLR7/8
clearance

Optimization Target: Dose adjustment guidelines for renally
impaired patients

13. Study Procedures & Methodology

Management Pathway: PK sampling schedule mapping

Education Protocol (HCP): Site-specific protocol training
regarding proper handling and processing of PK bio-samples

Education Protocol (Patient): Dietary and concomitant medication
restriction instructions prior to dosing

Quality Audit Mechanism: Data clarification checks via
centralized PK monitoring algorithms

14. Observation & Follow-Up Schedule

Total Duration: Up to 35 days per participant

Onsite Visit Cadence: Intensive monitoring Days 1-3, followed by
specific safety visits

Remote Monitoring Frequency: N/A (Predominantly onsite for Phase
1 PK parameters)

Checklist Review Interval: Routine site AE recording check

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				